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AI3403: Multi-Agent Systems (MAS) - Assignment #1

Rules for Submitting Assignment

- The assignment is worth 100 points containing 5 problems, each worth 20 points. You must present the entire work leading to their solution and also attach the associated code.
 - The assignment will be due by **12:00 (noon) on Friday, August 21, 2026** in the Google classroom. The deadline is strict and no extensions shall be awarded unless otherwise for unavoidable circumstances.
 - You should submit your assignment as **one** pdf document and the file should be named using roll number. For instance, if your roll number is **ai25btech11111**, then the pdf should be named as **ai25btech11111Assignment1.pdf**. Please also write your name in the first page of your assignment.
 - Students are strongly encouraged to submit their work in a neatly typed $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$ document with all supporting code attached to the same document. An example overleaf latex template can be found in this link: <https://www.overleaf.com/latex/templates/assignment-homework-solutions-template/nfjybpqbdrn>. If typing is not possible, hand-written submissions will be considered as long as they are legibly written and scanned properly with all the codes attached into one pdf.
 - Students can use any coding language of their preference such as Matlab, Python, Julia, C++, etc. All the associated codes for the assignment problems should be attached at the end of each respective problem in your assignment submission. You can also use Jupyter notebook, Google colab, etc to do your assignment and submit it as one pdf.
 - You can use internet, any books, lecture notes, etc. Discussing with other students is perfectly fine and in fact is strongly encouraged. But plagiarism (copying solutions from others without putting your hard work) will never be tolerated.
 - Usage of AI tools like ChatGPT/Claude/Gemini is permitted and must be limited only to understand the concepts taught in the lecture. If you use them in the assignment, you should explicitly declare where and to what extent you have used it. You must never ask AI Tools to solve the entire assignment problems for you. My aim is to educate you and not the AI tools!
 - There is also a bonus problem to ignite your creativity. Attempt it for your own learning and it *will not be graded*.
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Problem 1: Simulate the following tasks and give your inferences and all the associated codes.

1. For $N = 5$, generate and plot a random undirected graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$ using Erdos-Renyi model.
 - (a) Report the adjacency matrix, degree matrix, incidence matrix, and Laplacian matrix
 - (b) Compute and report the eigenvalues of Laplacian matrix. Report the algebraic connectivity λ_2 value. Explain the properties of the graph using the computed λ_2 quantity and mention why it is so important.
 - (c) Argue why high connectivity (i.e., each nodes having more connections) is a double-edged sword for agents in a network.
2. Assume the number of agents $N = 10^3$ and generate a random undirected connected graph (having spanning tree) $\mathcal{G}(\mathcal{V}, \mathcal{E})$ with N agents using Erdos-Renyi model and plot it.
 - (a) Highlight the node(s) with the highest number of neighbours with bright colour. Compute the mean degree, mean edges from the generated random graph and compare that with the theoretical value to see if it matches. Plot the diameter of the graph by colouring the corresponding edges.
 - (b) Pick any two nodes in random from the generated graph to compute the shortest path between them using any Shortest Path (SP) algorithm like Dijkstra algorithm and plot the shortest path between them in the graph.

Note: Both Matlab & Python have readily available package for plug and use, so that you need not write the code for the SP algorithm.

Problem 2: We denote the i^{th} component of a vector x at time $k \in \mathbb{N}$ as $(x_k)_i$. Consider a linear system given by $x_{t+1} = Ax_t, t = 1, \dots, T$, with $x_t \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$. Find the matrix $S \in \mathbb{R}^{n \times n}$ that depends upon the matrix A to measure the n numbers $\{(x_1)_1, (x_2)_2, \dots, (x_n)_n\}$, i.e., the i^{th} component of the state at time i , for $i = 1, \dots, n$ such that

$$\begin{bmatrix} (x_1)_1 \\ (x_2)_2 \\ \vdots \\ (x_n)_n \end{bmatrix} = Sx_1.$$

Problem 3: Let $A = \begin{bmatrix} B & I_n \\ 0_n & B \end{bmatrix}$ with $B \in \mathbb{R}^{n \times n}$. Compute the following.

1. Find e^{At}

2. Let $n = 2$ with $\alpha \in \mathbb{R}, \beta > 0$ and assume $B = \begin{bmatrix} 1 & \beta \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \alpha & 1 \\ 0 & \alpha \end{bmatrix} \begin{bmatrix} 1 & -\beta \\ 0 & 1 \end{bmatrix}$. Compute e^{At}

Problem 4: Given an interval $[a, b]$ with $a < b$ and a polynomial $p(x) = \sum_{k=0}^{n-1} c_{k+1}x^k$ with coefficients $c \in \mathbb{R}^n$, prove that both integral and derivative of the polynomial $p(x)$ is a linear function of its coefficients. That is, prove that $\exists \alpha \in \mathbb{R}^n, \exists \beta \in \mathbb{R}^n$ such that

$$\begin{aligned}\alpha^\top c &= \int_a^b p(x) dx \\ \beta^\top c &= p'(x) = \frac{dp(x)}{dx}\end{aligned}$$

Problem 5: Let $\{u_i \mid i \in \mathbb{N}_{\geq 1}\}$ be a sequence of vectors. Given $x_1 = 0$, let

$$x_t = \frac{1}{t-1} \sum_{k=1}^{t-1} u_k, \quad \forall t \in \mathbb{N}_{\geq 2}.$$

This represents a recursive mean computation of a collection of vectors by accessing them one by one. Find A, B matrices such that $x_{t+1} = Ax_t + Bu_t$ for $t = 1, 2, \dots$

Bonus Problem: Prove or disprove that the following sets are linear(vector) spaces.

1. Set of natural numbers, $\mathbb{N}$
2. Set of diagonal matrices
3. Set of bounded sequences, $\{\alpha[i], i \in \mathbb{N} : |\alpha[i]| < \infty\}$